2024

BOTANY — HONOURS

Paper : DSCC-4

(Plant Anatomy and Embryology)

Full Marks : 75

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer ***any six*** questions from the following : 2×6

   (a) Define Plastochron.

   (b) How Phelloderm is formed?

   (c) Define Apospory and Apogamy.

   (d) What is 'heartwood'?

   (e) Distinguish between leaf trace and leaf gap.

   (f) Distinguish between Apoplast and Symplast.

   (g) Define Apomixis.

   (h) What is Microgametogenesis?

   (i) Distinguish between fascicular and interfascicular cambium.

2. Answer ***any three*** questions from the following : 5×3

   (a) Discuss the types of stomata according to Stebbins and Khush (1961).

   (b) Write a note on the secondary growth found in monocot stem.

   (c) Give an overview of the ultrastructure of the cell wall of vascular plants.

   (d) Write a note on adaptive anatomical features of xerophytes.

   (e) Briefly describe three types of endosperm developments found in angiosperm.

3. Answer ***any four*** questions from the following :

   (a) Write a note on mechanical tissues found in different plant organs. Explain with suitable diagrams the principles governing their distribution in plants. 4+8

   (b) Define stele. Describe different types of stele with examples and diagrams. 1+6+2+3

   (c) Discuss the secondary growth in the stems of *Bignonia* and *Tecoma* with diagrams. Comment on their adaptive advantages. 4+4+4

(d) Diagrammatically represent the Korper-Kappe theory in relation to root apex organization. Write the adaptive anatomical features of halophytes. 8+4

(e) What do you mean by 'Dendrochronology'? Discuss, in brief, the different applications of plant anatomy in Systematics and Forensics. 2+5+5

(f) With suitable diagram describe the embryogenesis in *Capsella*. Describe, in brief, the tetrasporic type of embryo sac development. 8+4
